

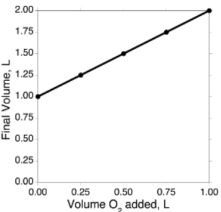
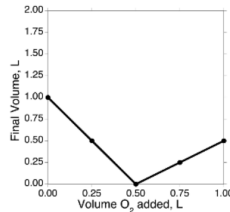
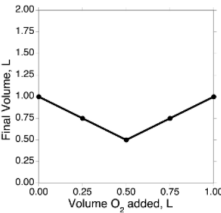
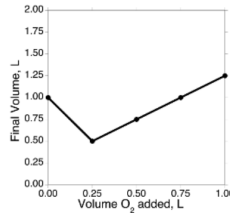
Stoichiometry/Solutions

- How many atoms are there in 0.300 mol of potassium nitrate, KNO_3 ($M = 101.11$)?
(A) 1.79×10^{21} (B) 1.81×10^{23}
(C) 5.42×10^{23} (D) 9.03×10^{23}
- A student isolates 22.0 g of ethyl bromide, $\text{C}_2\text{H}_5\text{Br}$ ($M = 108.96$), from the reaction of 34.0 g of ethyl alcohol ($M = 46.07$) with 59.0 g of phosphorus tribromide ($M = 270.67$) according to the balanced reaction below. What is the percent yield of the reaction?
 $3 \text{C}_2\text{H}_5\text{OH}(l) + \text{PBr}_3(l) \rightarrow 3 \text{C}_2\text{H}_5\text{Br}(l) + \text{H}_3\text{PO}_3(s)$
(A) 27.4% (B) 30.9% (C) 64.7% (D) 92.6%
- A 25.00 mL sample of aqueous HCl is titrated with 0.150 M NaOH , which requires adding 18.75 mL of the NaOH solution to reach the equivalence point. What is the concentration of chloride ion in the final solution?
(A) 0.0643 M (B) 0.0750 M
(C) 0.112 M (D) 0.200 M
- A colorless liquid hydrocarbon contains carbon and hydrogen in a 6.0 : 1 ratio by mass. Which is its molecular formula?
(A) CH (B) CH_2
(C) C_5H_{10} (D) C_6H_6
- Mixing equal volumes of 0.15 M $\text{Ba}(\text{NO}_3)_2$ and 0.20 M NaF causes the precipitation of insoluble barium fluoride. In which are the concentrations of ions in the supernatant listed in increasing order?
(A) $[\text{Ba}^{2+}] < [\text{F}^-] < [\text{NO}_3^-] < [\text{Na}^+]$
(B) $[\text{Ba}^{2+}] < [\text{F}^-] < [\text{Na}^+] < [\text{NO}_3^-]$
(C) $[\text{F}^-] < [\text{Ba}^{2+}] < [\text{NO}_3^-] < [\text{Na}^+]$
(D) $[\text{F}^-] < [\text{Ba}^{2+}] < [\text{Na}^+] < [\text{NO}_3^-]$
- A solution of which compound, at a concentration of 25 g L^{-1} , has the lowest freezing point?
(A) NaBrO_3 (B) KBrO_3
(C) Na_2SeO_4 (D) K_2SeO_4
- How many nitrogen atoms are present in 486 g of N_2O_5 ?
(A) 1.35×10^{24} (B) 2.71×10^{24}
(C) 5.42×10^{24} (D) 1.90×10^{25}
- What is the empirical formula of a compound that is 50% S and 50% O by mass?
(A) SO (B) S_2O
(C) SO_2 (D) S_2O_3
- How much chlorine is required to completely consume 13.5 g of aluminum to form aluminum chloride?
(A) 13.5 g (B) 26.6 g
(C) 40.5 g (D) 53.2 g
- A solution of 0.500 L of 0.150 M KI is mixed with 0.250 L of 0.200 M CaI_2 . What is the concentration of iodide in the final solution?
(A) 0.167 M (B) 0.175 M
(C) 0.233 M (D) 0.275 M
- The only source of copper in an ore sample is the mineral malachite, $\text{Cu}_2\text{CO}_3(\text{OH})_2$ ($M = 221.1$). A 1.00 kg sample of this ore is smelted to obtain 47.0 g of metallic copper. Assuming a complete recovery of the copper, what percentage by mass of the ore is malachite?
(A) 4.70% (B) 8.18%
(C) 16.4% (D) 32.8%
- Which aqueous solution has the highest boiling point when 10.00 g of the substance is dissolved in 100.0 g of water?
(A) H_2SO_4 (B) Li_2SO_4
(C) Na_2SO_4 (D) MgSO_4
- How many moles of ions are in 75.0 mL of a 1.20 M aqueous solution of sodium sulfate?
(A) 0.090 mol (B) 0.120 mol
(C) 0.180 mol (D) 0.270 mol
- When 3.00 g of $\text{NH}_3(g)$ ($M = 17.03$) reacts with 18.0 g of $\text{HCl}(g)$ ($M = 36.46$), what is the theoretical yield of $\text{NH}_4\text{Cl}(s)$?
(A) 6.42 g (B) 8.41 g
(C) 9.42 g (D) 21.0 g
- How many fluorine atoms are in 135 g of CF_2Cl_2 ?
(A) 3.36×10^{23} (B) 6.72×10^{23}
(C) 1.34×10^{24} (D) 4.28×10^{24}

Stoichiometry/Solutions

16. A 2024 US penny (which is a mixture of zinc and copper) is dissolved in dilute nitric acid, forming aqueous Cu^{2+} and Zn^{2+} ions. The resulting solution is treated with an excess of iodide ion, causing the Cu^{2+} to react to form $\text{CuI}(s)$ and $\text{I}_3^-(aq)$. Subsequent titration with thiosulfate reveals that 4.93×10^{-4} mol of $\text{I}_3^-(aq)$ was formed. What is the mass of copper in the penny?
- (A) 31.3 mg (B) 62.7 mg
(C) 125 mg (D) 188 mg
17. 1.22 g of benzoic acid ($\text{C}_7\text{H}_6\text{O}_2$, $M = 122.12$) is dissolved in water to a total volume of 150.0 mL. A 10.0 mL sample of this solution is then withdrawn and diluted to 500.0 mL. What is the concentration of benzoic acid in the final solution?
- (A) 0.00133 M (B) 0.00266 M
(C) 0.0200 M (D) 0.122 M
18. A 0.75 *m* aqueous solution of which substance has the lowest freezing point?
- (A) HF (B) HI
(C) H_3PO_4 (D) CH_3COOH
19. A 1.00 g sample of which compound will produce the greatest amount of carbon dioxide after complete combustion with excess oxygen?
- (A) CH_4 (B) C_3H_8 (C) C_6H_6 (D) C_8H_{18}
20. How many chlorine atoms are in 1.00 mL of CCl_4 ($M = 153.8$, density = 1.59 g mL^{-1})?
- (A) 1.56×10^{21} (B) 6.23×10^{21}
(C) 1.57×10^{22} (D) 2.49×10^{22}
21. What is the molality of Na^+ in a solution containing 3.00 g NaCl ($M = 58.4$), 9.00 g glucose ($M = 180.0$), and 168 g H_2O ($M = 18.0$)?
- (A) $5.50 \times 10^{-3} m$ (B) 0.285 *m*
(C) 0.306 *m* (D) 0.777 *m*
22. 25.0 mL of aqueous 0.20 M HCl , 10.0 mL of 0.25 M aqueous HNO_3 , and 10.0 mL of aqueous 0.30 M $\text{Ba}(\text{OH})_2$ are mixed to give a homogeneous solution. In which are the final concentrations of species ranked in order from highest to lowest?
- (A) $[\text{Cl}^-] > [\text{Ba}^{2+}] > [\text{NO}_3^-] > [\text{H}^+] > [\text{OH}^-]$
(B) $[\text{H}^+] > [\text{OH}^-] > [\text{Cl}^-] > [\text{Ba}^{2+}] > [\text{NO}_3^-]$
(C) $[\text{Cl}^-] > [\text{H}^+] > [\text{Ba}^{2+}] > [\text{NO}_3^-] > [\text{OH}^-]$
(D) $[\text{OH}^-] > [\text{H}^+] > [\text{Ba}^{2+}] > [\text{NO}_3^-] > [\text{Cl}^-]$
23. Acetanilide ($\text{C}_6\text{H}_5\text{NHCOCH}_3$, $M = 135.2$) can be prepared from aniline ($\text{C}_6\text{H}_5\text{NH}_2$, $M = 93.1$) and acetic anhydride ($(\text{CH}_3\text{CO})_2\text{O}$, $M = 102.1$) according to the balanced equation below.
- $$\text{C}_6\text{H}_5\text{NH}_2 + (\text{CH}_3\text{CO})_2\text{O} \rightarrow \text{C}_6\text{H}_5\text{NHCOCH}_3 + \text{CH}_3\text{COOH}$$
- If 10.0 g acetanilide is isolated from the reaction of 10.0 g aniline with 10.0 g acetic anhydride, what is the percent yield of the reaction?
- (A) 50.0% (B) 68.9% (C) 69.3% (D) 75.5%
24. 10.0 g of a metal chloride salt is dissolved in 90.0 g H_2O . This solution has a vapor pressure that is 8.6% lower than that of pure water at the same temperature. What is the identity of the metal?
- (A) Li (B) Na (C) Mg (D) Ca
25. A 1.00-g sample of which compound will produce the greatest amount of carbon dioxide after complete combustion with excess oxygen?
- (A) CH_4 (B) C_3H_6 (C) C_6H_{14} (D) C_8H_{18}
26. Rhenium (Re) consists of two stable isotopes, ^{185}Re and ^{187}Re . The average atomic mass of Re is 186.2 amu. What is the mole percent of ^{185}Re in rhenium?
- (A) 40% (B) 50%
(C) 60% (D) 62%
27. A 10.00 g sample of a soluble barium salt is treated with an excess of sodium sulfate to precipitate 8.93 g BaSO_4 ($M = 233.4$). Which barium salt is it?
- (A) BaCl_2 ($M = 208.2$)
(B) $\text{Ba}(\text{O}_2\text{CH})_2$ ($M = 227.3$)
(C) $\text{Ba}(\text{NO}_3)_2$ ($M = 261.3$)
(D) BaBr_2 ($M = 297.1$)
28. How much $\text{Sr}(\text{OH})_2 \cdot 8 \text{H}_2\text{O}$ ($M = 265.76$) is needed to prepare 150.0 mL of solution in which $[\text{OH}^-] = 0.100 \text{ M}$?
- (A) 1.99 g (B) 3.98 g
(C) 5.98 g (D) 7.98 g

Stoichiometry/Solutions

29. Which procedure produces a 0.2 M solution of Na_2SO_4 ?
- (A) Mixing 500 mL of 0.4 M NaOH with 500 mL of 0.2 M Na_2SO_4
 (B) Mixing 300 mL of 1.2 M NaOH with 600 mL of 0.3 M H_2SO_4
 (C) Dissolving 0.2 mol solid Na_2SO_4 in 500 mL water
 (D) Diluting 400 mL of 0.3 M Na_2SO_4 to a final volume of 1.0 L
30. 10.0 g of an alkali metal chloride salt is dissolved in 90.0 g H_2O . This solution has a vapor pressure that is 3.2% lower than that of pure water at the same temperature. What is the salt?
- (A) LiCl (B) NaCl (C) KCl (D) RbCl
31. Which potassium salt is 28.9% chlorine by mass?
- (A) KCl (B) KClO (C) KClO_2 (D) KClO_3
32. How many sulfur atoms are in 3.00 g of iron pyrite, FeS_2 ($M = 120.0$)?
- (A) 7.53×10^{21} (B) 1.51×10^{22}
 (C) 3.01×10^{22} (D) 6.02×10^{23}
33. A gas mixture contains 50 mol% H_2 and 50 mol% He. 1.00-L samples of this gas mixture are mixed with variable volumes of O_2 (at 0 °C and 1 atm). A spark is introduced to allow the mixture to undergo complete combustion. The final volume is measured at 0 °C and 1 atm. Which graph best depicts the final volume as a function of the volume of added O_2 ?
- (A)  (B) 
- (C)  (D) 
34. What is the concentration of nitrate ions in 40.0 mL of a 0.25 M solution of copper(II) nitrate?
- (A) 0.010 M (B) 0.020 M
 (C) 0.050 M (D) 0.50 M
35. 1.000 g of a transition metal carbonate, MCO_3 , is heated to produce the metal oxide and 0.383 g of carbon dioxide ($M = 44.01$). What is the identity of the metal M?
- (A) Mn (B) Ni (C) Cu (D) Zn
36. 14.0 mL of 0.53 M barium nitrate solution and 16.0 mL of 0.44 M sodium sulfate solution are mixed. Which ion is LEAST abundant in solution?
- (A) Ba^{2+} (B) NO_3^- (C) Na^+ (D) SO_4^{2-}
37. At 120 °C and 1 atm pressure, 1.00 L of methane, CH_4 , reacts completely with excess oxygen to form carbon dioxide and water. What volumes of the two products are produced at this pressure and temperature?
- (A) 1.00 L CO_2 and 2.00 L H_2O
 (B) 1.00 L CO_2 and 4.00 L H_2O
 (C) 2.00 L CO_2 and 2.00 L H_2O
 (D) 2.00 L CO_2 and 4.00 L H_2O
38. Polypropylene is made by polymerizing propene, C_3H_6 ($M = 42.1$). How many molecules of propene must be polymerized to make 3.50 g polypropylene?
- (A) 1.43×10^{22} (B) 5.01×10^{22}
 (C) 6.02×10^{23} (D) 2.11×10^{24}
39. An organic compound contains only carbon, hydrogen, nitrogen, and oxygen. It is 61.71% C, 4.03% H, and 16.00% N by mass. What is its empirical formula?
- (A) $\text{C}_5\text{H}_4\text{NO}$ (B) $\text{C}_9\text{H}_7\text{N}_2\text{O}_2$
 (C) $\text{C}_{10}\text{H}_8\text{N}_2\text{O}$ (D) $\text{C}_{11}\text{H}_8\text{NO}_2$
40. 100.0 mL of 0.500 M CaBr_2 and 50.0 mL of 1.00 M NaBr are mixed. What is the concentration of bromide ion in the resulting solution?
- (A) 0.500 M (B) 0.667 M
 (C) 0.750 M (D) 1.00 M
41. 1.00 g of hydrated potassium carbonate, $\text{K}_2\text{CO}_3 \cdot n\text{H}_2\text{O}$, is heated to 250 °C to give 0.836 g anhydrous K_2CO_3 . What is the value of n ?
- (A) 0.16 (B) 1.0 (C) 1.5 (D) 2.0

Stoichiometry/Solutions

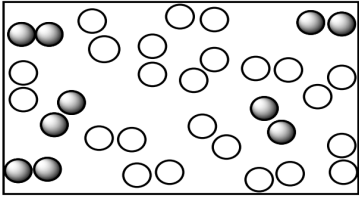
42. The concentration of an aqueous solution of a nonvolatile, monoprotic acid is measured first by freezing point depression and then by boiling point elevation. The solution is found to be $0.93\ m$ by freezing point depression and to be $0.82\ m$ by boiling point elevation. Which is the best explanation for this discrepancy?
- (A) Ionization of the acid is markedly exothermic.
(B) The solute associates partially into dimers at lower temperatures.
(C) The volume of the solution is greater at higher temperatures.
(D) The boiling point elevation constant for water is smaller than its freezing point depression constant.
43. A 2.50 L sample of butane gas (C_4H_{10}), measured at $22.0^\circ C$ and 1.20 atm pressure, is combusted completely and the carbon dioxide gas collected at the same pressure and temperature. What volume of CO_2 is produced?
- (A) 2.50 L (B) 9.00 L
(C) 10.0 L (D) 22.5 L
44. How many hydrogen atoms are in 2.50 g of pharmacolite, $CaHAsO_4 \cdot 2H_2O$ ($M = 216.0$)?
- (A) 6.97×10^{21} (B) 2.09×10^{22}
(C) 2.79×10^{22} (D) 3.48×10^{22}
45. Methanol is produced industrially by catalytic hydrogenation of carbon monoxide according to the following equation:
- $$CO(g) + 2 H_2(g) \rightarrow CH_3OH(l)$$
- If the yield of the reaction is 40%, what volume of CO (measured at STP) would be needed to produce 1.0×10^6 kg CH_3OH ?
- (A) 2.8×10^8 L (B) 7.0×10^8 L
(C) 1.7×10^9 L (D) 2.1×10^9 L
46. What volume of 18.0 M sulfuric acid must be diluted to 250.0 mL to afford a 0.55 M solution of sulfuric acid?
- (A) 3.1 mL (B) 4.5 mL (C) 7.6 mL (D) 31 mL
47. What is the concentration of H_2O_2 in a solution that is 30.0% by mass hydrogen peroxide and has a density of $1.11\ g\ cm^{-3}$?
- (A) 9.79 M (B) 12.6 M
(C) 18.5 M (D) 32.6 M
48. A 5.00 g mixture of potassium sulfide and potassium chloride contains 2.80 g potassium. What percentage by mass of the mixture is potassium sulfide?
- (A) 13.8% (B) 19.2% (C) 44.0% (D) 96.1%
49. Which hydrocarbon is 84.1% carbon by mass?
- (A) CH_4 (B) C_2H_6 (C) C_4H_{10} (D) C_8H_{18}
50. Lithium hydrogen carbonate, $LiHCO_3$, decomposes when heated to form Li_2O , CO_2 , and H_2O . How many moles of H_2O are formed when 0.50 mol $LiHCO_3$ decomposes?
- (A) 0.25 mol (B) 0.50 mol
(C) 0.75 mol (D) 1.0 mol
51. A vessel contains 66 g CO_2 and 16 g O_2 at a pressure of 10.0 atm. What is the partial pressure of CO_2 ?
- (A) 5.0 atm (B) 6.0 atm
(C) 7.5 atm (D) 8.0 atm
52. First-generation automobile airbags were inflated by the decomposition of sodium azide ($M = 65.02$):
- $$2 NaN_3(s) \rightarrow 2 Na(s) + 3 N_2(g)$$
- What mass of sodium azide would be required to inflate a 16.0 L airbag to a pressure of 1.20 atm at $17^\circ C$?
- (A) 34.9 g (B) 52.4 g (C) 78.6 g (D) 157 g
53. Which of the following combinations will produce a 0.4 M NaCl solution?
- (A) Mixing 500 mL of 0.4 M NaOH with 500 mL of 0.4 M HCl
(B) Mixing 300 mL of 1.2 M NaOH with 600 mL of 0.6 M HCl
(C) Mixing 500 mL of 0.4 M NaCl solution with 500 mL water
(D) Diluting 400 mL of 0.6 M NaCl with water to a final volume of 1.0 L
54. Which solute, when dissolved in 1 kg H_2O , will give the solution with the highest boiling point?
- (A) 50 g ethylene glycol, $HOCH_2CH_2OH$
(B) 50 g hydroxyacetic acid, $HOCH_2COOH$
(C) 50 g glycerol, $HOCH_2CHOHCH_2OH$
(D) 50 g glyoxal, $HCOCHO$

Stoichiometry/Solutions

55. How many oxygen atoms are in 225 g O₂?
(A) 4.23×10^{24} (B) 6.84×10^{24}
(C) 8.47×10^{24} (D) 1.69×10^{25}
56. A 0.300 M solution of HCl is prepared by adding some 1.50 M HCl to a 500 mL volumetric flask and diluting to the mark with deionized water. What volume of 1.50 M HCl must be added?
(A) 100. mL (B) 150. mL
(C) 225. mL (D) 250. mL
57. Copper(I) oxide, Cu₂O, is reduced to metallic copper by heating in a stream of hydrogen gas. What mass of water is produced when 10.00 g copper is formed?
(A) 1.259 g (B) 1.417 g
(C) 2.835 g (D) 5.670 g
58. The mineral enargite is 48.41% Cu, 19.02% As, and 32.57% S by mass. What is the empirical formula of enargite?
(A) CuAsS (B) Cu₂AsS₂
(C) Cu₃AsS₄ (D) Cu₄AsS₃
59. A 5.0 g sample of calcium nitrate (Ca(NO₃)₂, $M = 164$) contaminated with silica (SiO₂, $M = 60.1$) is found to contain 1.0 g calcium. What is the mass percent purity of calcium nitrate in the sample?
(A) 20% (B) 24% (C) 73% (D) 82%
60. A solution of 20.0 g of which hydrated salt dissolved in 200 g H₂O will have the lowest freezing point?
(A) CuSO₄ • 5 H₂O ($M = 250$)
(B) NiSO₄ • 6 H₂O ($M = 263$)
(C) MgSO₄ • 7 H₂O ($M = 246$)
(D) Na₂SO₄ • 10 H₂O ($M = 286$)
61. Glyoxal consists of 41.4% C, 3.5% H, and 55.1% O by mass. What is the empirical formula of glyoxal?
(A) CHO (B) CH₂O
(C) CH₂O₂ (D) C₁₂H₁₆O₁₆
62. How many hydroxide ions are in 2.5 mol Mg(OH)₂?
(A) 3.0×10^{23} (B) 6.0×10^{23}
(C) 1.5×10^{24} (D) 3.0×10^{24}
63. In a sample consisting of 1.00 mol NaBr and 0.300 mol KI, what is the mass percent of iodine?
(A) 24.9% (B) 32.6% (C) 47.2% (D) 83.1%
64. What is the concentration of chloride ions in a solution formed by mixing 150. mL of a 1.50 M NaCl solution with 250. mL of a 0.750 M MgCl₂ solution?
(A) 0.563 M (B) 1.03 M
(C) 1.50 M (D) 2.25 M
65. Barium chloride reacts with sodium sulfate according to the following equation:
$$\text{BaCl}_2(aq) + \text{Na}_2\text{SO}_4(aq) \rightarrow \text{BaSO}_4(s) + 2 \text{NaCl}(aq)$$

A student mixes a solution containing 10.0 g BaCl₂ ($M = 208.2$) with a solution containing 10.0 g Na₂SO₄ ($M = 142.1$) and obtains 12.0 g BaSO₄ ($M = 233.2$). What is the percent yield of this reaction?
(A) 60.0%
(B) 73.1%
(C) 93.3%
(D) The isolated barium sulfate is most likely wet, since the yield would otherwise be greater than 100%.
66. Which aqueous solution has the highest boiling point?
(A) 1.0 *m* acetic acid, CH₃COOH
(B) 1.0 *m* sulfuric acid, H₂SO₄
(C) 1.0 *m* phosphoric acid, H₃PO₄
(D) 1.0 *m* glucose, C₆H₁₂O₆
67. A 20.0 g sample of mercury(II) oxide (HgO, $M = 216.6$) is heated strongly, causing it to decompose to metallic Hg and O₂ gas. What volume of O₂ gas is produced (measured at STP)?
(A) 1.03 L (B) 2.07 L (C) 4.14 L (D) 14.0 L
68. When 30.0 mL of 0.10 M AgNO₃ is added to 30.0 mL of 0.10 M NaCl, aqueous NaNO₃ and solid AgCl are formed. How much solid AgCl is produced?
(A) 0.0030 mol (B) 0.0060 mol
(C) 0.030 mol (D) 0.060 mol
69. How much Sr(OH)₂ • 8 H₂O ($M = 265.76$) is needed to prepare 250.0 mL of solution in which [OH⁻] = 0.100 M?
(A) 3.32 g (B) 6.64 g (C) 9.97 g (D) 13.3 g

Stoichiometry/Solutions

70. A 10.00 g sample of a compound containing only carbon, hydrogen, and oxygen forms 23.98 g CO₂ and 4.91 g H₂O upon complete combustion. What is the empirical formula of the compound?
- (A) C₂H₂O (B) C₃H₃O (C) C₆H₃O₂ (D) C₆H₆O
71. Which of the following is a nonelectrolyte in aqueous solution?
- (A) H₂SO₄ (B) NaC₂H₃O₂
(C) K₂CO₃ (D) CH₂O
72. An aqueous solution of potassium sulfate (K₂SO₄) has a freezing point of -2.24 °C. What is its molality? ($K_f = 1.86\text{ }^{\circ}\text{C}\cdot m^{-1}$)
- (A) 0.401 *m* (B) 0.602 *m*
(C) 1.20 *m* (D) 4.17 *m*
73. Benzene, C₆H₆, reacts with oxygen, O₂, to form CO₂ and H₂O. How much O₂ is required for the complete combustion of 1.0 mol C₆H₆?
- (A) 6.0 mol (B) 7.5 mol
(C) 9.0 mol (D) 12 mol
74. A 10.00 g sample of a soluble barium salt is treated with an excess of sodium sulfate to precipitate 11.21 g BaSO₄ ($M = 233.4$). Which barium salt is it?
- (A) BaCl₂ ($M = 208.2$)
(B) Ba(O₂CH)₂ ($M = 227.3$)
(C) Ba(NO₃)₂ ($M = 261.3$)
(D) BaBr₂ ($M = 297.1$)
75. What is the concentration of nitrate ion in a 425 mL solution containing 32.0 g of Mg(NO₃)₂ ($M = 148.3$)?
- (A) 0.216 M (B) 0.432 M
(C) 0.508 M (D) 1.02 M
76. The formula for terbium phosphate is TbPO₄. The formula for terbium sulfate is
- (A) Tb₂SO₄ (B) TbSO₄
(C) Tb₂(SO₄)₃ (D) Tb(SO₄)₂
77. A 25.0 mL sample of 0.15 M silver nitrate, AgNO₃, is reacted with a 3.58 g sample of calcium chloride, CaCl₂ ($M = 111.0$). Which of the following statements is true?
- (A) Silver nitrate is the limiting reactant and calcium nitrate precipitates.
(B) Silver nitrate is the limiting reactant and silver chloride precipitates.
(C) Calcium chloride is the limiting reactant and calcium nitrate precipitates.
(D) Calcium chloride is the limiting reactant and silver chloride precipitates.
78. Which aqueous solution exhibits the largest freezing point depression?
- (A) 1.0 *m* KBr (B) 0.75 *m* C₆H₁₂O₆
(C) 0.5 *m* MgCl₂ (D) 0.25 *m* Ga₂(SO₄)₃
79. Which compound contains the highest percentage of magnesium by mass?
- (A) MgNH₄PO₄ (B) Mg(H₂PO₄)₂
(C) Mg₂P₄O₇ (D) Mg₃(PO₄)₂
80. 
- In the diagram above the paired open spheres represent H₂ molecules and the paired solid spheres represent N₂ molecules. When the molecules in the box react to form the maximum possible amount of ammonia (NH₃) molecules, what is the limiting reactant and how many molecules of NH₃ can be formed?
- (A) N₂ is limiting. 5 molecules of NH₃ can be formed.
(B) N₂ is limiting. 10 molecules of NH₃ can be formed.
(C) H₂ is limiting. 8 molecules of NH₃ can be formed.
(D) H₂ is limiting. 12 molecules of NH₃ can be formed.
81. Vanillin, C₈H₈O₃ ($M = 152\text{ g/mol}$), is the molecule responsible for the vanilla flavor in food. How many oxygen atoms are present in a 45.0 mg sample of vanillin?
- (A) 1.78×10^{20} (B) 5.35×10^{20}
(C) 1.78×10^{23} (D) 5.35×10^{23}

Stoichiometry/Solutions

82. What is the molarity of sodium ions in a solution prepared by diluting 250. mL of 0.550 M Na_2SO_4 to 1.25 L?
(A) 0.110 M (B) 0.138 M
(C) 0.220 M (D) 0.275 M
83. The solubility of $\text{K}_2\text{Cr}_2\text{O}_7$ in water is 125 g/L at 20 °C. A solution is prepared at 20 °C that contains 6.0 grams of $\text{K}_2\text{Cr}_2\text{O}_7$ in 50. mL of water. This solution is
(A) dilute. (B) saturated.
(C) supersaturated. (D) unsaturated.
84. Which of the following is a weak electrolyte in aqueous solution?
(A) HF (B) NaF (C) HCl (D) KCl
85. When this equation is balanced using the smallest possible integers, what is the sum of the coefficients?
 $\text{---}(\text{NH}_4)_3\text{PO}_4(\text{aq}) + \text{---}\text{CaCl}_2(\text{aq}) \rightarrow$
 $\text{---}\text{Ca}_3(\text{PO}_4)_2(\text{s}) + \text{---}\text{NH}_4\text{Cl}(\text{aq})$
(A) 8 (B) 9 (C) 11 (D) 12
86. A solution of KNO_3 in water is prepared for which the following data have been obtained:
masses of solute and solvent
molar masses of solute and solvent
Which of these quantitative descriptions of the solution can be determined?
I. molarity II. molality III. density of solution
(A) I. only (B) II. only
(C) I. and II. only (D) I., II. and III.
87. What mass of the compound CrO_3 ($M = 100.0$) contains 4.5×10^{23} oxygen atoms?
(A) 2.25 g (B) 12.0 g (C) 25.0 g (D) 75.0 g
88. An 18.5 g sample of tin ($M = 118.7$) combines with 10.0 g of sulfur ($M = 32.07$) to form a compound. What is the empirical formula of this compound?
(A) SnS (B) SnS_2 (C) Sn_2S (D) Sn_2S_3
89. A mixture is prepared by adding 50.0 mL of 0.200 M NaOH to 75.0 mL of 0.100 M NaOH. What is the $[\text{OH}^-]$ in the mixture?
(A) 0.0600 M (B) 0.0800 M
(C) 0.140 M (D) 0.233 M
90. What mass of NaHCO_3 ($M = 84.0$) is required to completely neutralize 25.0 mL of 0.125 M H_2SO_4 ?
(A) 0.131 g (B) 0.262 g
(C) 0.525 g (D) 1.05 g
91. How many atoms are in 4.0×10^{-5} grams of Al?
(A) 8.9×10^{17} (B) 2.4×10^{19}
(C) 6.5×10^{20} (D) 2.0×10^{22}
92. How many moles of sulfate ions are in 100 mL of a solution of 0.0020 M $\text{Fe}_2(\text{SO}_4)_3$?
(A) 2.0×10^{-4} (B) 6.0×10^{-4}
(C) 2.0×10^{-1} (D) 6.0×10^{-1}
93. The value of which concentration unit for a solution changes with temperature?
(A) molarity (B) molality
(C) mole fraction (D) mass percentage
94. How many moles of water are produced by the complete combustion of 14.4 g of C_5H_{12} ?
$$\text{C}_5\text{H}_{12} + 8\text{O}_2 \rightarrow 5\text{CO}_2 + 6\text{H}_2\text{O}$$

(A) 0.200 (B) 0.600 (C) 1.20 (D) 2.40
95. A solution of $\text{Ba}(\text{OH})_2$ is standardized with potassium acid phthalate (abbreviated KHP), $\text{KHC}_8\text{H}_4\text{O}_4$ ($M = 204$). If 1.530 g of KHP is titrated with 34.50 mL of the $\text{Ba}(\text{OH})_2$ solution, what is the molarity of $\text{Ba}(\text{OH})_2$?
(A) 0.0217 M (B) 0.0435 M
(C) 0.109 M (D) 0.217 M
96. 30.0 mL of 0.10 M $\text{Ca}(\text{NO}_3)_2$ and 15.0 mL of 0.20 M Na_3PO_4 solutions are mixed. After the reaction is complete, which of these ions has the lowest concentration in the final solution?
(A) Na^+ (B) NO_3^- (C) Ca^{2+} (D) PO_4^{3-}
97. A 65.25 g sample of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ ($M = 249.7$) is dissolved in enough water to make 0.800 L of solution. What volume of this solution must be diluted with water to make 1.00 L of 0.100 M CuSO_4 ?
(A) 3.27 mL (B) 81.6 mL
(C) 209 mL (D) 306 mL

Stoichiometry/Solutions

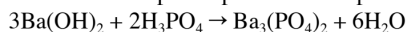
98. How many moles of O_2 are required for the complete combustion of 2.2 g of C_3H_8 to form CO_2 and H_2O ?
 (A) 0.050 (B) 0.15 (C) 0.25 (D) 0.50
99. A typical polyethylene bag from a grocery store weighs 12.4 g. How many molecules of ethylene, C_2H_4 , must be polymerized to make such a bag?
 (A) 1.36×10^{24} (B) 6.02×10^{23}
 (C) 5.33×10^{23} (D) 2.67×10^{23}
100. In the titration of a monoprotic acid with a solution of sodium hydroxide of known concentration, what quantities are equal at the equivalence point?
 (A) the concentrations of hydroxide and hydronium ions
 (B) the number of moles of hydroxide ion added and the number of moles of hydronium ion initially present
 (C) the volume of sodium hydroxide solution added and the volume of acid solution initially present
 (D) the number of moles of hydroxide ion added and the number of moles of monoprotic acid initially present
101. When a nonvolatile solute is dissolved in a volatile solvent, which characteristic is greater for the solution than for the solvent?
 (A) boiling point (B) freezing point
 (C) rate of evaporation (D) vapor pressure
102. What is the concentration of nitrate ion in a solution made by mixing 100 mL of a 0.200 M HNO_3 solution with 200 mL of a 0.100 M solution of $Mg(NO_3)_2$?
 (A) 0.133 M (B) 0.150 M
 (C) 0.167 M (D) 0.200 M
103. What is the coefficient for O_2 when the following reaction
 $__As_2S_3 + __O_2 \rightarrow __As_2O_3 + __SO_2$
 is correctly balanced with the smallest integer coefficients?
 (A) 5 (B) 6 (C) 8 (D) 9
104. Which compound contains the highest percentage of nitrogen by mass?
 (A) NH_2OH ($M = 33.0$) (B) NH_4NO_2 ($M = 64.1$)
 (C) N_2O_3 ($M = 76.0$) (D) $NH_4NH_2CO_2$ ($M = 78.1$)
105. How many neutrons are in 0.025 mol of the isotope $^{54}_{24}Cr$?
 (A) 1.5×10^{22} (B) 3.6×10^{23}
 (C) 4.5×10^{23} (D) 8.1×10^{23}
106. Magnesium chloride dissolves in water to form
 (A) hydrated $MgCl_2$ molecules
 (B) hydrated Mg^{2+} ions and hydrated Cl^- ions
 (C) hydrated Mg^{2+} ions and hydrated Cl_2^{2-} ions
 (D) hydrated Mg atoms and hydrated Cl_2 molecules
107. Enzymes convert glucose ($M = 180.2$) to ethanol ($M = 46.1$) according to the equation

$$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$$

 What is the maximum mass of ethanol that can be made from 15.5 kg of glucose?
 (A) 0.256 kg (B) 0.512 kg
 (C) 3.96 kg (D) 7.93 kg
108. Commercial vinegar is a 5.00% by mass aqueous solution of acetic acid, CH_3CO_2H ($M = 60.0$). What is the molarity of acetic acid in vinegar? [density of vinegar = 1.00 g/mL]
 (A) 0.833 M (B) 1.00 M
 (C) 1.20 M (D) 3.00 M
109. The mass of one atom of an element is 1.71×10^{-22} g. What is the atomic mass of this element in $g \cdot mol^{-1}$?
 (A) 101 (B) 103 (C) 105 (D) 107
110. What is the percent by mass of nitrogen in ammonium carbonate, $(NH_4)_2CO_3$?
 (A) 14.53% (B) 27.83%
 (C) 29.16% (D) 33.34%
111. How many moles of water will be produced from the complete combustion of 4.4 g of C_3H_8 ?
 (A) 0.10 (B) 0.25 (C) 0.40 (D) 0.80
112. A 10.0 g sample of an oxide of copper forms metallic copper and 1.26 g of water when heated in a stream of hydrogen. What is the mass percent of copper in this oxide?
 (A) 11.2% (B) 66.6% (C) 79.9% (D) 88.8%
113. A 49.9 g sample of barium hydroxide octahydrate, $Ba(OH)_2 \cdot 8H_2O$, is dissolved in water and the solution is diluted to give a final volume of 2.50 L. What is the concentration of the hydroxide ion in this solution?
- | Molar Mass / $g \cdot mol^{-1}$ | |
|---------------------------------|-----|
| $Ba(OH)_2 \cdot 8H_2O$ | 315 |
- (A) 0.0634 M (B) 0.127 M
 (C) 0.190 M (D) 0.634 M

Stoichiometry/Solutions

114. What volume (in mL) of 0.0500 M phosphoric acid is needed to titrate completely 25.0 mL of 0.150 M barium hydroxide solution to a phenolphthalein end point?



(A) 50.0 (B) 75.0 (C) 100. (D) 150.

115. A 1.0 gram sample of which substance contains the largest number of molecules?

(A) COCl_2 (B) CS_2 (C) CH_3Cl (D) $\text{C}_2\text{H}_2\text{F}_2$

116. How many moles of ions are present in 250 mL of a 4.4 M solution of sodium sulfate?

(A) 1.1 (B) 2.2 (C) 3.3 (D) 13

117. How many moles of oxygen gas are produced by the decomposition of 245 g of potassium chlorate?

Molar Mass / $\text{g}\cdot\text{mol}^{-1}$	
KClO_3	122.6



(A) 1.50 (B) 2.00 (C) 2.50 (D) 3.00

118. A 50.0 mL solution of 0.150 M HCl is mixed with 25.0 mL of 0.400 M HCl. What is the HCl concentration in the final solution? (Assume volumes are additive.)

(A) 0.0175 (B) 0.233 (C) 0.275 (D) 0.550

119. For the reaction: $2\text{X} + 3\text{Y} \rightarrow 3\text{Z}$, the combination of 2.00 moles of X with 2.00 moles of Y produces 1.75 moles of Z. What is the percent yield of this reaction?

(A) 43.8% (B) 58.3% (C) 66.7% (D) 87.5%

120. For which property is the value greater for a solution of a nonvolatile solute than for the pure solvent?

(A) boiling point (B) freezing point
(C) triple point (D) vapor pressure

121. A saturated aqueous solution of sucrose, $\text{C}_{12}\text{H}_{22}\text{O}_{11}$, contains 525 g of sucrose (molar mass 342) per 100. g of water. What is the $\text{C}_{12}\text{H}_{22}\text{O}_{11}/\text{H}_2\text{O}$ molecular ratio in this solution?

(A) 5.25/1 (B) 1.54/1
(C) 1/1 (D) 0.276/1

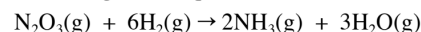
122. The mineral beryl contains 5.03% beryllium by mass and contains three beryllium atoms per formula unit. Determine the formula mass of beryl.

(A) 950 g/mol (B) 537 g/mol
(C) 270 g/mol (D) 179 g/mol

123. A 100. mL portion of 0.250 M calcium nitrate solution is mixed with 400. mL of 0.100 M nitric acid solution. What is the final concentration of the nitrate ion?

(A) 0.180 M (B) 0.130 M
(C) 0.0800 M (D) 0.0500 M

124. According to the equation,



how many moles of $\text{NH}_3(\text{g})$ could be formed from the reaction of 0.22 mol of $\text{N}_2\text{O}_3(\text{g})$ with 0.87 mol of $\text{H}_2(\text{g})$?

(A) 0.29 mol (B) 0.44 mol
(C) 0.73 mol (D) 1.1 mol

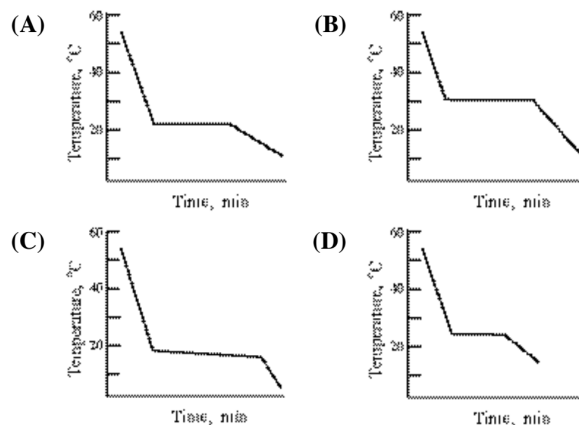
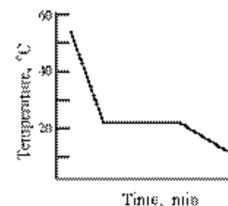
125. Water can be decomposed by the passage of an electric current according to the equation



How many moles of $\text{H}_2(\text{g})$ can be produced from the passage of 4.8×10^{21} electrons?

(A) 2.00×10^{-3} (B) 4.0×10^{-3}
(C) 8.0×10^{-3} (D) 1.6×10^{-2}

126. This diagram represents the behavior of a pure solvent upon cooling. Which of the diagrams below best represents the cooling curve of a solution in that solvent upon cooling? (Assume that all diagrams are drawn to the same scale.)



Stoichiometry/Solutions

127. Which is the net ionic equation for the reaction when 0.10 M solutions of silver nitrate and sodium sulfide are mixed?

(A) $\text{Ag}^+(\text{aq}) + \text{S}^{2-}(\text{aq}) \rightarrow \text{AgS}(\text{s})$
 (B) $\text{Ag}^{2+}(\text{aq}) + \text{S}^{2-}(\text{aq}) \rightarrow \text{AgS}(\text{s})$
 (C) $\text{Ag}^{2+}(\text{aq}) + \text{S}^{2-}(\text{aq}) \rightarrow \text{AgS}_2(\text{s})$
 (D) $\text{Ag}^+(\text{aq}) + \text{S}^{2-}(\text{aq}) \rightarrow \text{Ag}_2\text{S}(\text{s})$

128. Magnetite, which has the formula Fe_3O_4 , is comprised of iron(II) oxide and iron(III) oxide. What is the ratio of iron(II) ions to iron(III) ions in magnetite?

(A) 1:1 (B) 1:2 (C) 2:3 (D) 3:2

129. $\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$
 What is the total mass, in grams, of products when 2.20 g of propane is burned in excess oxygen?

(A) 2.20 (B) 3.60 (C) 6.60 (D) 10.2

130. What volume, in mL, of concentrated sulfuric acid (18.0 M H_2SO_4) is needed to prepare 2.50 L of a 1.00 M solution?

(A) 7.20 (B) 14.4 (C) 69.4 (D) 139

131. The bromide impurity in a 2.00 g sample of a metal nitrate is precipitated as silver bromide. If 6.40 mL of 0.200 M AgNO_3 solution is required, what is the mass percentage of bromide in the sample?

(A) 1.28 (B) 2.56 (C) 5.11 (D) 9.15

132. Under which conditions is the solubility of oxygen gas in water the greatest?

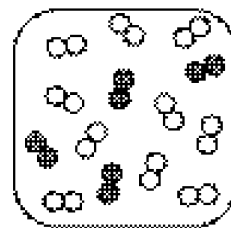
Pressure Temperature

(A) high high
 (B) high low
 (C) low high
 (D) low low

133. What is the mass of one molecule of water in grams?

(A) 18 (B) 1.1×10^{-21}
 (C) 3.0×10^{-23} (D) 1.7×10^{-24}

134. If nitrogen atoms are represented as filled circles and oxygen atoms as open circles, how much NO_2 can be prepared from the mixture shown?



(A) 4 molecules (B) 5 molecules
 (C) 6 molecules (D) 8 molecules

135. A mineral containing iron(II) sulfide but no other sulfides is treated with excess hydrochloric acid to produce hydrogen sulfide. If a 3.15 g sample of the mineral yields 448 mL of hydrogen sulfide gas (measured at 0 °C and 760 mm Hg), what is the mass percentage of iron(II) sulfide in the sample?

(A) 20.4 (B) 35.5 (C) 55.8 (D) 71.0

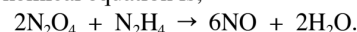
136. Naturally occurring thallium consists of two stable isotopes, Tl-203 and Tl-205 (atomic masses = 203.0 and 205.0, respectively) and has an average atomic mass of 204.4. What is the percentage of Tl-205?

(A) 14.0% (B) 30.0% (C) 50.0 % (D) 70.0 %

137. What is the maximum mass (in grams) of NO that could be obtained from

Molar Mass ($\text{g}\cdot\text{mol}^{-1}$)	
N_2O_4	92.0
N_2H_4	32.0

15.5 g of N_2O_4 and 4.68 g of N_2H_4 when they react? The balanced chemical equation is;



(A) 4.38 (B) 5.04 (C) 15.2 (D) 26.2

138. What volume of 0.108 M H_2SO_4 is required to neutralize 25.0 mL of 0.145 M KOH?

(A) 16.8 mL (B) 33.6 mL
 (C) 37.2 mL (D) 67.1 mL

139. For which compound are the empirical and molecular formulas the same?

(A) $\text{C}_6\text{H}_5\text{COOH}$ (B) $\text{C}_6\text{H}_4(\text{COOH})_2$
 (C) HOCCOOH (D) CH_3COOH

140. What volume of liquid A has the same mass as 80.0 cm^3 of liquid B?

Density (g/cm^3)	
Liquid A	0.660
Liquid B	1.59

(A) 40.0 cm^3 (B) 97.0 cm^3
 (C) 160. cm^3 (D) 193 cm^3

Stoichiometry/Solutions

141. How many water molecules are in a 0.10 g sample of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (MM = 249.7)?
 (A) 1.2×10^{21} (B) 2.4×10^{21}
 (C) 2.4×10^{22} (D) 1.2×10^{23}
142. Acetylene, C_2H_2 , reacts with O_2 to produce CO_2 and H_2O . What is the $\text{O}_2/\text{C}_2\text{H}_2$ ratio in the balanced equation?
 (A) 2/1 (B) 3/2 (C) 5/2 (D) 3/1
143. $\text{Mg}(\text{OH})_2$ in the form of Milk of Magnesia is used to neutralize excess stomach acid.

Molar Mass (g/mol)	
$\text{Mg}(\text{OH})_2$	58.33

 How many moles of stomach acid can be neutralized by 1.00 g of $\text{Mg}(\text{OH})_2$?
 (A) 0.0171 (B) 0.0343 (C) 0.686 (D) 1.25
144. A 25.00 mL sample of 0.1050 M H_2SO_4 is titrated with a NaOH solution of unknown concentration. The phenolphthalein endpoint was reached when 17.23 mL of the NaOH solution had been added. What is the concentration of the NaOH?
 (A) 0.07617 M (B) 0.1447 M
 (C) 0.1524 M (D) 0.3047 M
145. Selenium (Se) is similar to sulfur in its properties and francium (Fr) is an alkali metal. What is the formula for francium selenite?
 (A) FrSeO_2 (B) Fr_2SeO_4
 (C) Fr_2SeO_3 (D) $\text{Fr}_2\text{Se}_2\text{O}_3$
146. Calculate the mass percentage of nitrogen in hydrazinium sulfate ($\text{N}_2\text{H}_5)_2\text{SO}_4$.

Molar mass, g·mol ⁻¹	
$(\text{N}_2\text{H}_5)_2\text{SO}_4$	162.2

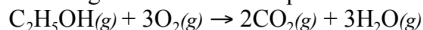
 (A) 10.8 (B) 17.3 (C) 34.5 (D) 51.2
147. How many ozone molecules are in 3.20 g of O_3 ?
 (A) 4.0×10^{22} (B) 6.0×10^{22}
 (C) 1.2×10^{23} (D) 6.0×10^{23}
148. Acetylene, C_2H_2 , reacts with oxygen according to the unbalanced equation:
 $\text{C}_2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$
 What is the $\text{O}_2/\text{C}_2\text{H}_2$ ratio when this equation is correctly balanced?
 (A) 2/1 (B) 3/1 (C) 4/1 (D) 5/2
149. Silicon carbide, SiC, is produced by heating SiO_2 and C to high temperatures according to the equation:
 $\text{SiO}_2(\text{s}) + 3\text{C}(\text{s}) \rightarrow \text{SiC}(\text{s}) + 2\text{CO}(\text{g})$
 How many grams of SiC could be formed by reacting 2.00 g of SiO_2 and 2.00 g of C?
 (A) 1.33 (B) 2.26 (C) 3.59 (D) 4.00
150. A 7.66 g sample of hydrated sodium sulfate, $\text{Na}_2\text{SO}_4 \cdot x\text{H}_2\text{O}$, forms 4.06 g of anhydrous Na_2SO_4 . What is the value of x?

Molar mass, g·mol ⁻¹	
Na_2SO_4	142

 (A) 0.2 (B) 3.6 (C) 5 (D) 7
151. Silver metal reacts with nitric acid according to the equation:
 $3\text{Ag}(\text{s}) + 4\text{HNO}_3(\text{aq}) \rightarrow 3\text{AgNO}_3(\text{aq}) + \text{NO}(\text{g}) + 2\text{H}_2\text{O}(\text{l})$
 What volume of 1.15 M $\text{HNO}_3(\text{aq})$ is required to react with 0.784 g of silver?
 (A) 4.74 mL (B) 6.32 mL
 (C) 8.43 mL (D) 25.3 mL
152. Which solute produces the highest boiling point in a 0.15 m aqueous solution?
 (A) CaCl_2 (B) NaBr (C) CuSO_4 (D) CH_3OH
153. An oxide of manganese contains 2.29 g of manganese per gram of oxygen. What is the empirical formula of this compound?
 (A) MnO (B) MnO_2
 (C) Mn_2O_3 (D) MnO_3
154. A heterogeneous system is produced when 0.040 moles of solid NaCl is added to 0.10 L of 0.10 M $\text{Pb}(\text{NO}_3)_2$. Which ion is present in the aqueous phase at the highest concentration?
 (A) Cl^- (B) NO_3^-
 (C) Pb^{2+} (D) Na^+
155. Which expression gives the fraction by mass of nitrogen in ammonium dihydrogen phosphate?
 (A) 14 / 115 (B) 28 / 115
 (C) 28 / 132 (D) 14 / 210

Stoichiometry/Solutions

156. Ethanol burns in excess oxygen to form $\text{CO}_2(g)$ and $\text{H}_2\text{O}(g)$ according to this balanced equation.



What value is closest to the volume of $\text{CO}_2(g)$, measured at 200K and 1 atm, produced from the combustion of 0.25 mol of $\text{C}_2\text{H}_5\text{OH}(g)$?

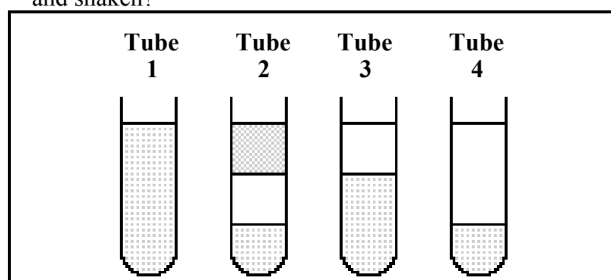
- (A) 5 L (B) 8 L
(C) 10 L (D) 15 L

157. Adipic acid, $\text{HOOC}(\text{CH}_2)_4\text{COOH}$, is used in making nylon. What is the total number of atoms in 1.0 g of adipic acid?

Molar Mass, $\text{g} \cdot \text{mol}^{-1}$	
adipic acid	146.26

- (A) 20 (B) 4.1×10^{21}
(C) 8.2×10^{22} (D) 7.2×10^{24}

158. Hexane, C_6H_{14} , is immiscible with water and ethanol. Water and ethanol are miscible. C_6H_{14} has the lowest density. Which diagram represents the results when equal volumes of these three liquids are placed in a test tube and shaken?



- (A) 1 (B) 2 (C) 3 (D) 4

159. $5\text{H}_2\text{C}_2\text{O}_4(aq) + 2\text{MnO}_4^-(aq) + 6\text{H}^+(aq) \rightarrow 2\text{Mn}^{2+}(aq) + 10\text{CO}_2(g) + 8\text{H}_2\text{O}(l)$

Oxalic acid, $\text{H}_2\text{C}_2\text{O}_4$, reacts with permanganate ion according to the balanced equation above. How many mL of 0.0154 M KMnO_4 solution are required to react with 25.0 mL of 0.0208 M $\text{H}_2\text{C}_2\text{O}_4$ solution?

- (A) 13.5 mL (B) 18.5 mL
(C) 33.8 mL (D) 84.4 mL

160. A chloride salt of rhenium contains 63.6% Re by mass. What is its empirical formula?

- (A) ReCl (B) ReCl_2
(C) ReCl_3 (D) ReCl_5

161. The amount of chloride ion in a water sample is to be determined by adding excess silver nitrate. If

Molar Mass, $\text{g} \cdot \text{mol}^{-1}$	
AgNO_3	169.91
AgCl	143.25

1.0 g of silver chloride is precipitated, what mass of chloride ion is in the original sample?

- (A) 0.25 g (B) 0.34 g
(C) 0.50 g (D) 0.75 g

162. $_\text{Sn}^{2+}(aq) + _\text{MnO}_4^-(aq) + _\text{H}^+(aq) \rightarrow _\text{Sn}^{4+}(aq) + _\text{Mn}^{2+}(aq) + _\text{H}_2\text{O}(l)$

When this equation for the reaction of $\text{Sn}^{2+}(aq)$ and $\text{MnO}_4^-(aq)$ is balanced correctly, what is the ratio, $\text{Sn}^{2+} / \text{MnO}_4^-$?

- (A) 1/1 (B) 1/2 (C) 2/1 (D) 5/2

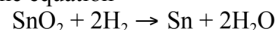
163. How many millimoles of methane, CH_4 , are present in 6.4 g of this gas?

- (A) 0.40 (B) 4.0
(C) 40. (D) 4.0×10^2

164. A 1.50 mL sample of a sulfuric acid solution from an automobile storage battery is titrated with 1.47 M sodium hydroxide solution to a phenolphthalein endpoint, requiring 23.70 mL. What is the molarity of the sulfuric acid solution?

- (A) 23.2 M (B) 11.6 M
(C) 6.30 M (D) 0.181 M

165. According to the equation



What volume of hydrogen, measured at 1 atm and 273 K, is required to react with 2.00 g of SnO_2 ?

- (A) 0.00133 L (B) 0.00265 L
(C) 0.297 L (D) 0.595 L

166. If 1.50 g of $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ were heated to drive off the water of hydration, how much anhydrous $\text{H}_2\text{C}_2\text{O}_4$ would remain?

- (A) 0.34 g (B) 0.92 g
(C) 1.07 g (D) 1.50 g

167. How many H atoms are in 3.4 g of $\text{C}_{12}\text{H}_{22}\text{O}_{11}$?

- (A) 6.0×10^{23} (B) 1.3×10^{23}
(C) 3.8×10^{22} (D) 6.0×10^{21}

Stoichiometry/Solutions

168. How many mL of 8.00 M HCl are needed to prepare 150. mL of a 1.60 M HCl solution?

(A) 30.0 mL (B) 24.0 mL
(C) 18.8 mL (D) 12.0 mL

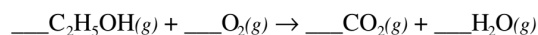
169. Analysis of a compound known to contain only Mg, P, and O gives this analysis.

21.8% Mg 27.7% P 50.3% O

What is its empirical formula?

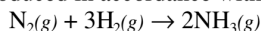
(A) MgPO_2 (B) MgPO_3
(C) $\text{Mg}_2\text{P}_2\text{O}_7$ (D) $\text{Mg}_3\text{P}_2\text{O}_8$

170. The reaction of ethanol, $\text{C}_2\text{H}_5\text{OH}$, with oxygen is a popular classroom demonstration. Balance the equation to find the number of moles of gaseous products formed per mole of ethanol.



(A) 2 (B) 3 (C) 4 (D) 5

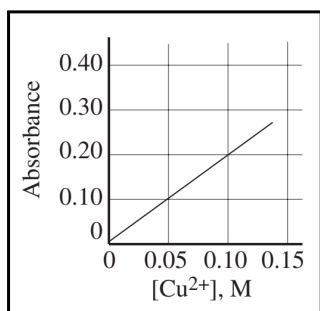
171. Ammonia is produced in accordance with this equation.



In a particular experiment, 0.25 mol of NH_3 is formed when 0.5 mol of N_2 is reacted with 0.5 mol of H_2 . What is the percent yield?

(A) 75% (B) 50%
(C) 33% (D) 25%

172. A 20.0 mL sample of a Cu^{2+} solution was diluted to 250.0 mL. A portion of this solution was found to have an absorbance of 0.15 under the same conditions that were used to generate the given absorbance vs. $[\text{Cu}^{2+}]$ graph. What was the concentration of Cu^{2+} ions in the original sample?



(A) 0.0060 M (B) 0.075 M
(C) 0.30 M (D) 0.94 M